

Introduction to Big Data

Reading Papers

Reading Papers

- How to read an engineering research paper by Bill Griswold and Gail Murphy
 - Just search on the web for the sources for this paper

Research Papers

- Style
 - Terse style due to page limits
 - Typical target audience: researchers in field
 - Intended audience is rarely you! ☹
- Reading as a budding researcher
 - May need to read additional papers such as cited papers, tutorial papers, and textbooks
 - Become an “active” reader
 - Become a skeptical reader
 - Read multiple times with different foci

Research Paper Types (1)

- Tutorial papers
 - Pages usually not limited or limit is high
 - Peer-reviewed over extended time (quality!)
 - Most often dated but useful for newbies
 - ACM Computing Surveys, conferences
- Journal papers
 - Pages usually not limited or limit is high
 - Peer-reviewed over extended time (quality?)
 - Often dated but serve as archival record
 - TODS, VLDB Journal, Data Engineering Bulletin

Research Paper Types (2)

- Conference papers
 - Strict page limits
 - Usually peer-reviewed over short time period
 - Reasonably current and correct
 - SIGMOD/PODS, VLDB, ICDE, CIDR
- Workshop papers
 - Often have lower page limits
 - Often peer-reviewed but less quality control
 - Provide access to preliminary work
 - Middleware, DOA, workshops in conferences

Research Paper Types (3)

- Technical reports
 - No page limits
 - Internal (department) publications
 - Rarely peer-reviewed so may have “bugs”
 - Best source of details

Industry Papers

- White papers, manuals, tutorials and similar documents
- Webcasts/podcasts
 - See videos on the course page
- Suggestion
 - Look for products that claim to work in the area of the research

Again be skeptical, very very skeptical ☹️

Questions To Ask While Reading

- What are or is ...
 - The motivations for this work?
 - The proposed solution?
 - The evaluation for the solution?
 - The existing alternatives?
 - The paper's contributions (if any)?
 - The anticipated future work?



Multiple Readings: Initial Pass

- Try to understand ...
 - Motivation and contributions
- Focus on high-level understanding of ...
 - Solution and evaluation criteria
- Definitely cover the following ...
 - Abstract, introduction, related work, and conclusions

Subsequent Passes (Note the Plural!)

Solution

- If paper has an algorithm try it on some examples
- If paper has an architecture diagram, try to trace some execution paths
- Mark up paper — annotations are useful for re-reading

Evaluation

- If paper presents a proof step through it
- If paper presents empirical data and results, check to see these really make sense

Focus on understanding solution or evaluation